

The business case for a new process of creating high value products from nickel laterites

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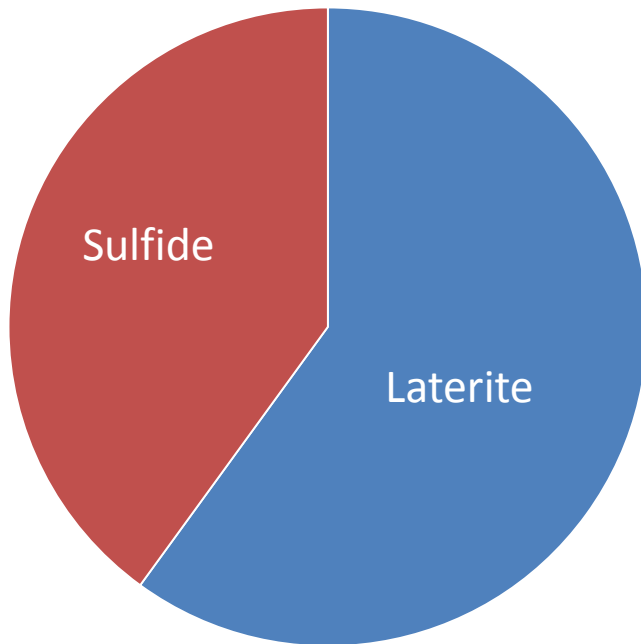
New Caledonia Nickel Conference

3 July 2013

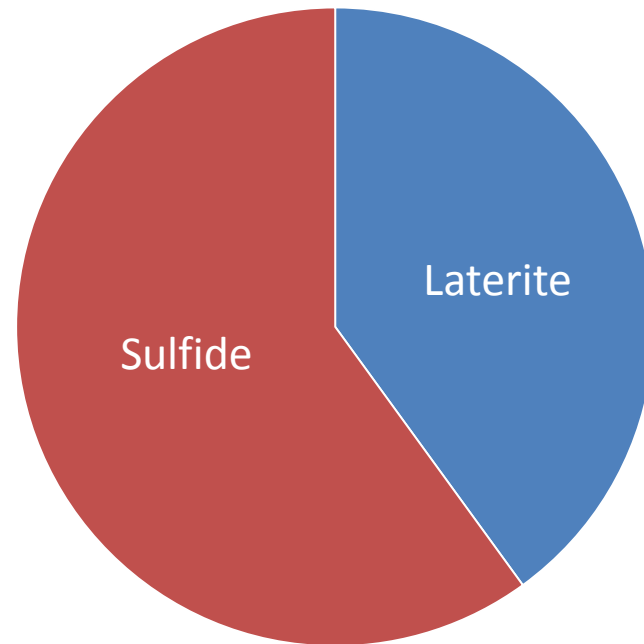


Low grade nickel laterites are becoming more important

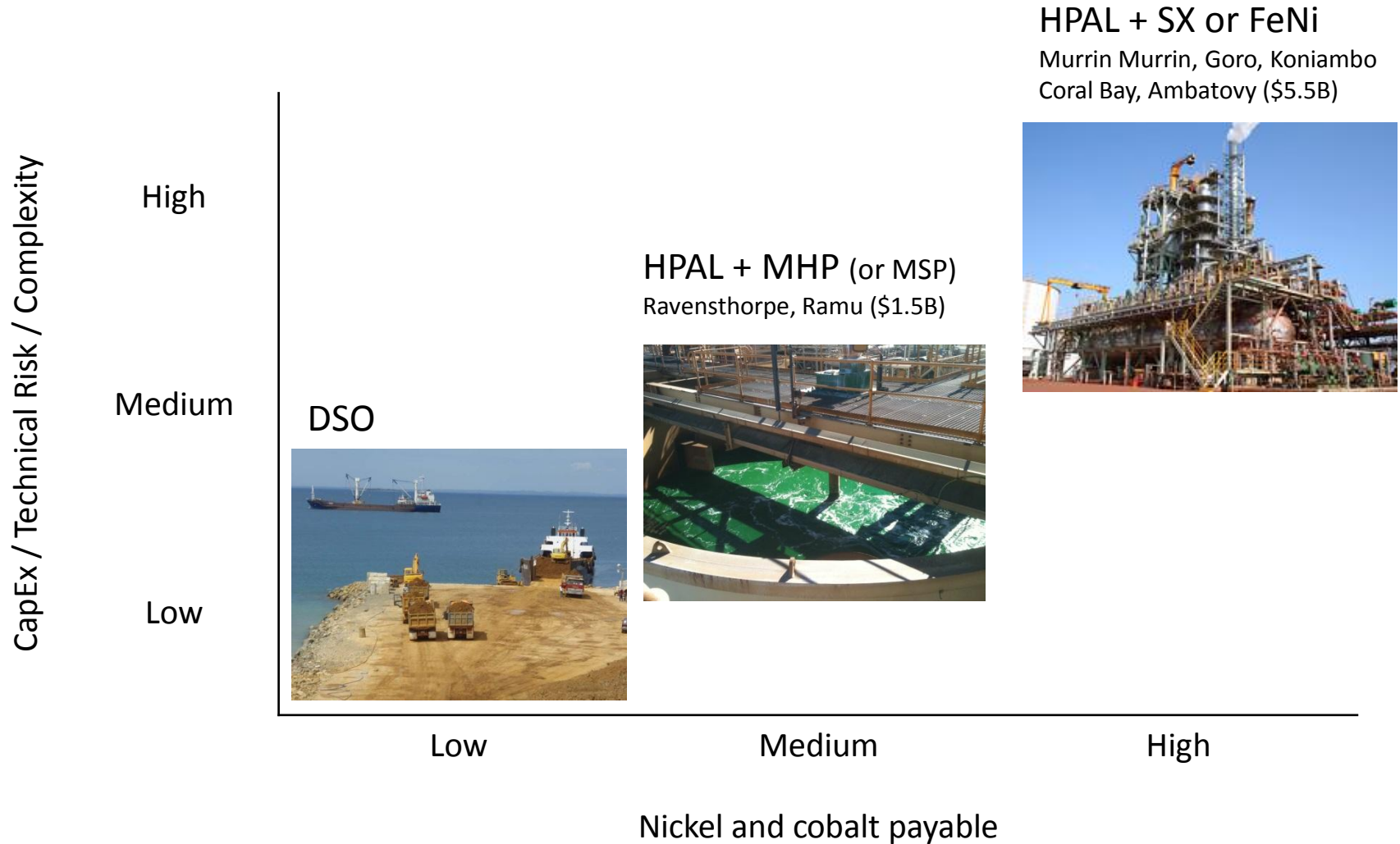
Reserves



Production



Laterite resource owners have a number of options

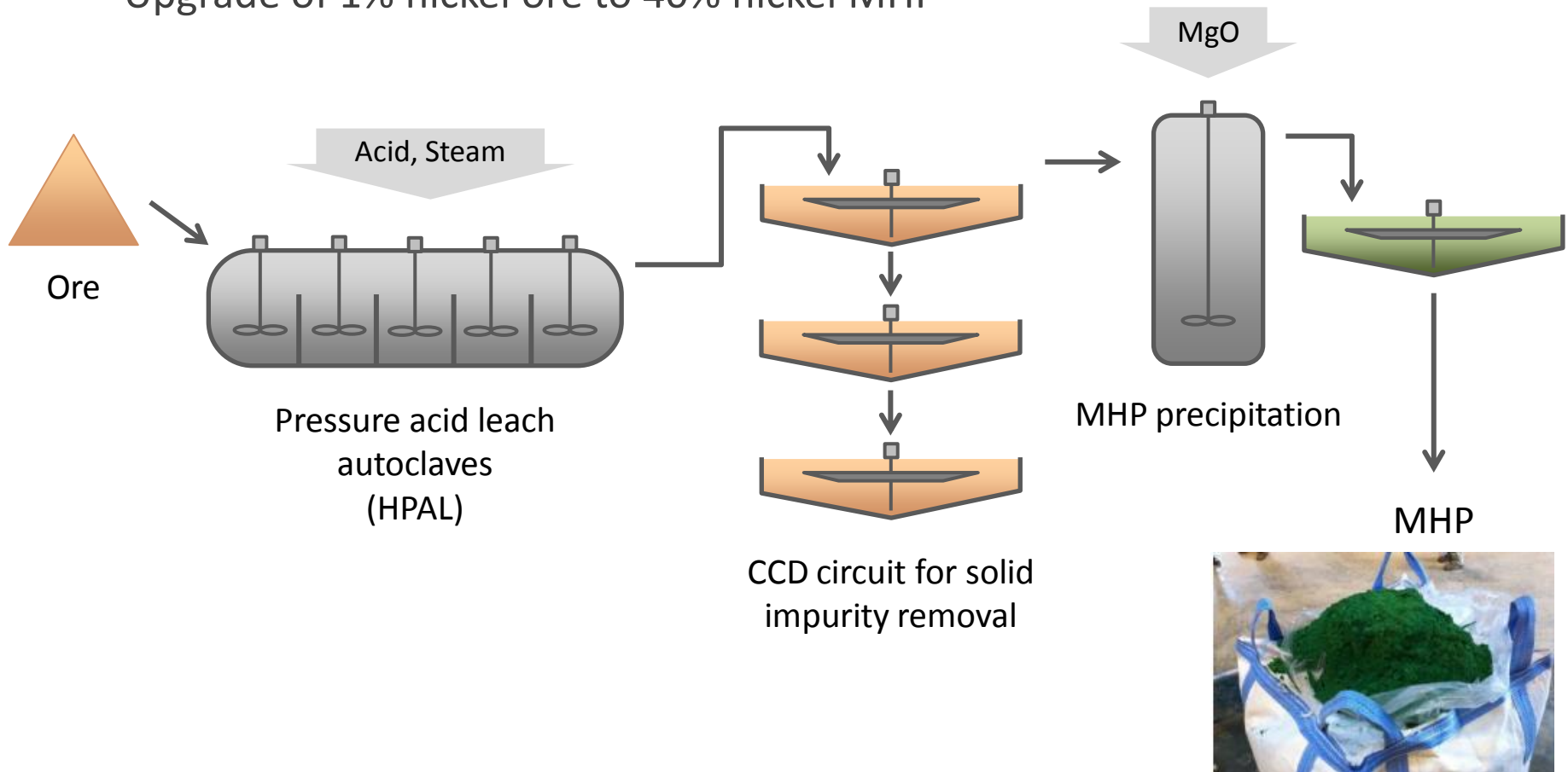


Direct shipping of ore

- Relatively low value for the miner
 - 15-20% of nickel value for 1.8% nickel ore
 - Typically shipped to China for NPI (Indonesia, Philippines)
- NPI is a major source of primary nickel
 - 30% of China's primary nickel consumption (Hatch estimate)
- Moving towards being dependent on high grade laterite ore
 - Producers prefer > 1.5% nickel grade
- Ore export is under threat in many countries
 - Indonesia to potentially ban exports of raw minerals in 2014
 - DRC copper concentrate export ban

Producing MHP as an intermediate material

- Mixed (nickel and cobalt) hydroxide precipitate
- Upgrade of 1% nickel ore to 40% nickel MHP



MHP is a low cost, marketable product

- Reduced capital requirements compared to final metal production
- Simple operation / Low technical risk
- Currently produced and sold by
 - First Quantum's Ravensthorpe (Australia) at 31,000-35,000 tpa of nickel as MHP¹
 - MCC's Ramu (PNG) shipped 2,000t of nickel as MHP Q1 2013²
- Under study for a number of projects
 - Agata (Mindoro/TVI), SCONI (Metallica), Mambare (Regency), Dutwa (African Eagle), Kalgoorlie (Heron), Wingellina (Metals X), Gladstone (GPNL), Northmet (Polymet), Mesaba (Teck)

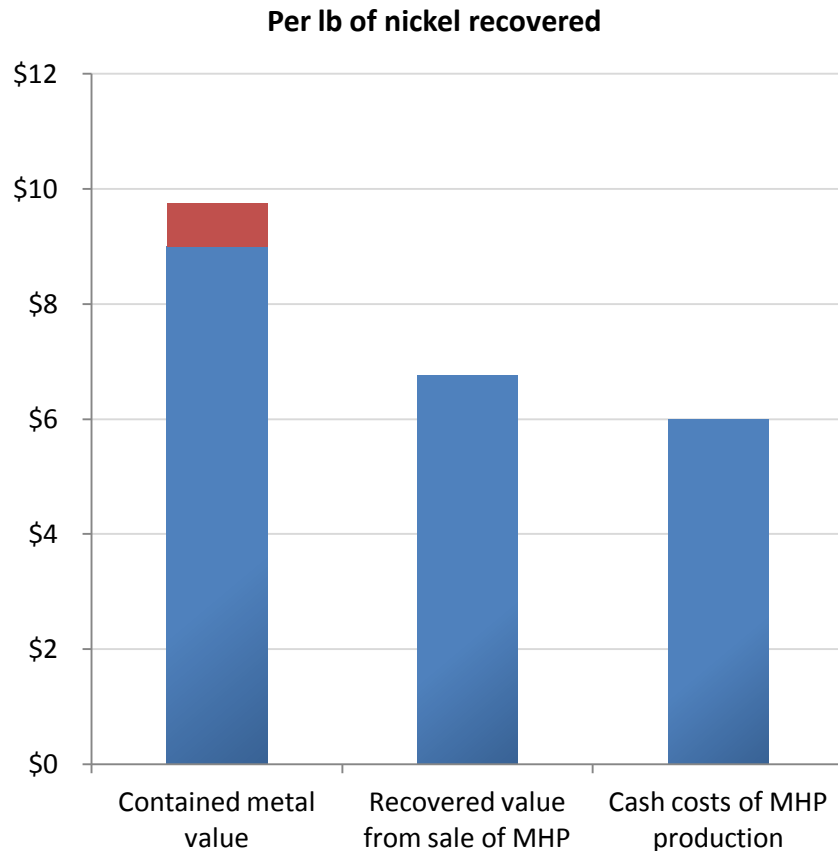
1. First Quantum operational outlook for Ravensthorpe in 2013 (Q1 2013 Financial Statement)
2. Highlands Pacific March 2013 Quarterly Report

But, relatively low payable on metal content

- MHP valued by the market at 70-80% of contained nickel value¹
- No or little value for contained cobalt
- Limited number of refiners
 - QNi's Palmer Nickel and Cobalt Refinery (Australia)
 - Jinchuan, Sichuan, Huaze (China)
 - One site in India, One site in Brazil
- Refiners have difficulty with some MHP
 - Aging
 - Impurities

1. First Quantum is obtaining 77% payable for Ravensthorpe MHP (Q1 2013 Financial Statement)

MHP production results in low profitability



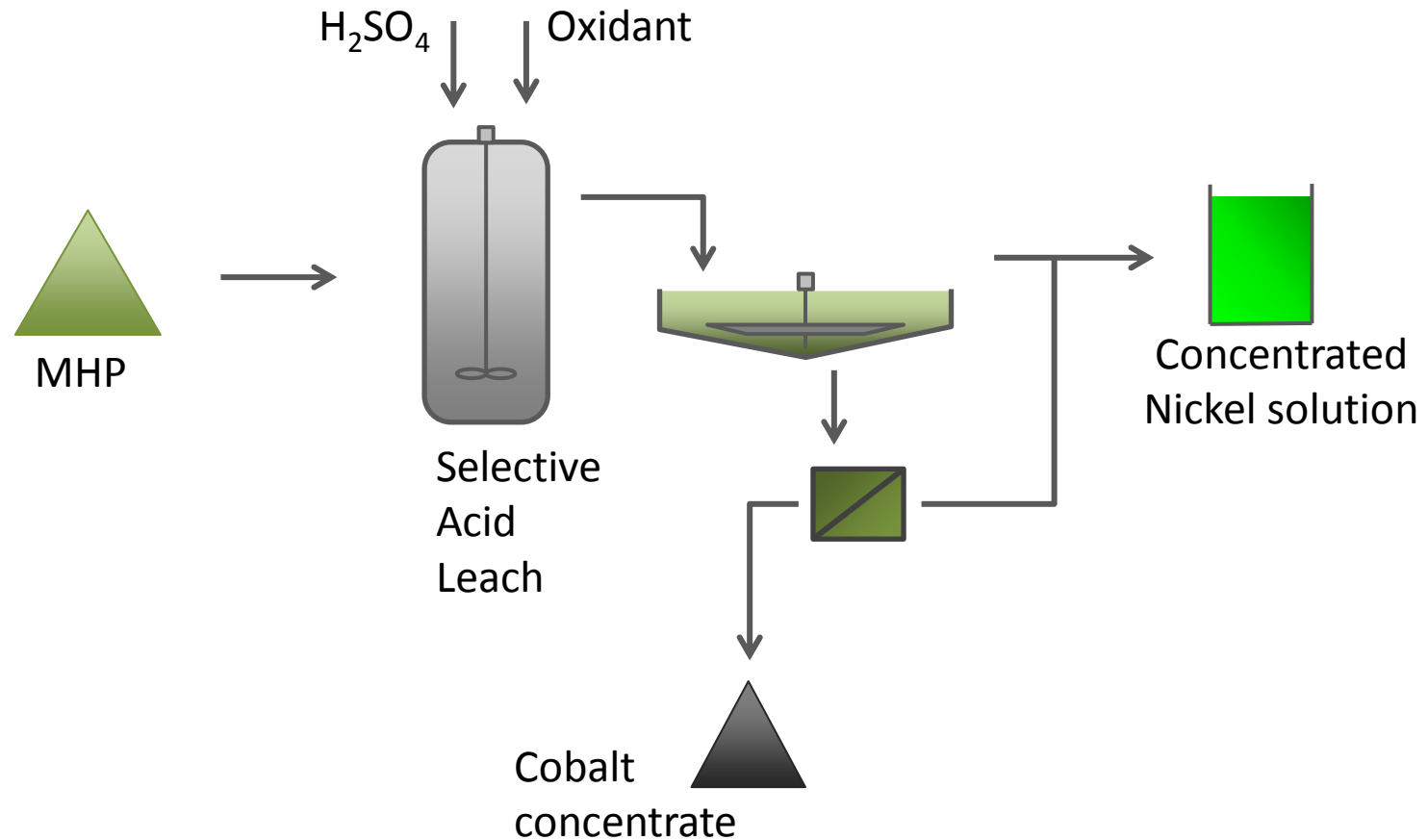
Mine life	20 years
Annual capacity	50,000 tpa of Ni 2,500 tpa of Co
Annual revenue ¹	\$745M
Cash costs	\$6 / lb recovered Ni ²
Annual operating costs	\$660M
Annual gross profit	\$82M
Capital cost	\$1,500M
Payback period	19 years
NPV (8%)	-\$690M
IRR	1%
Ni price for +’ve NPV³	\$9.85

1. Long term metal prices of \$9/lb nickel, \$15/lb cobalt, MHP sold at 75% of contained nickel value
2. C1 cash costs approximated from First Quantum’s Ravensthorpe operation
3. Assuming no change in cash costs

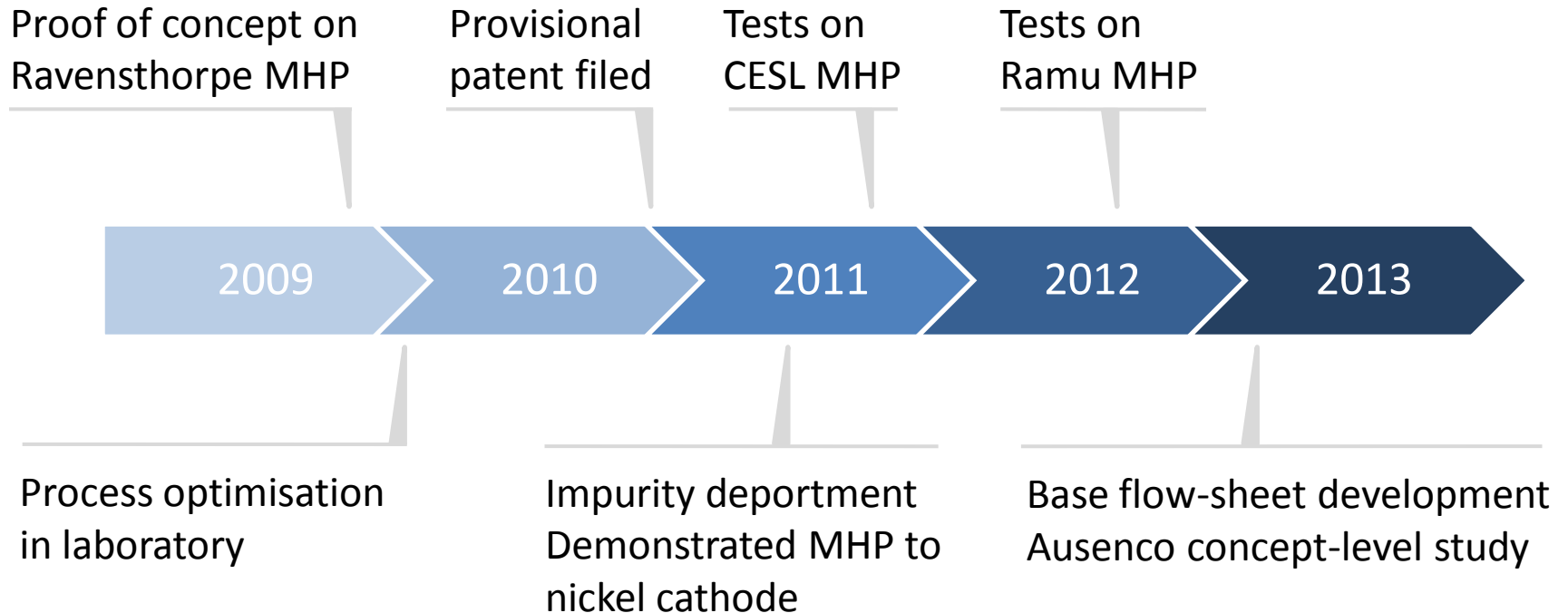
A number of technical options exist for the MHP refining

- Ammonia re-leach followed by solvent extraction
 - QNi at Palmer Nickel and Cobalt refinery
 - Cawse in Australia
- Acid re-leach followed by solvent extraction or ion exchange
 - Specified in two recent patents
- **Selective acid re-leach**
 - Developed by The University of Queensland
 - Robust, single-step process to leach and separate nickel from cobalt

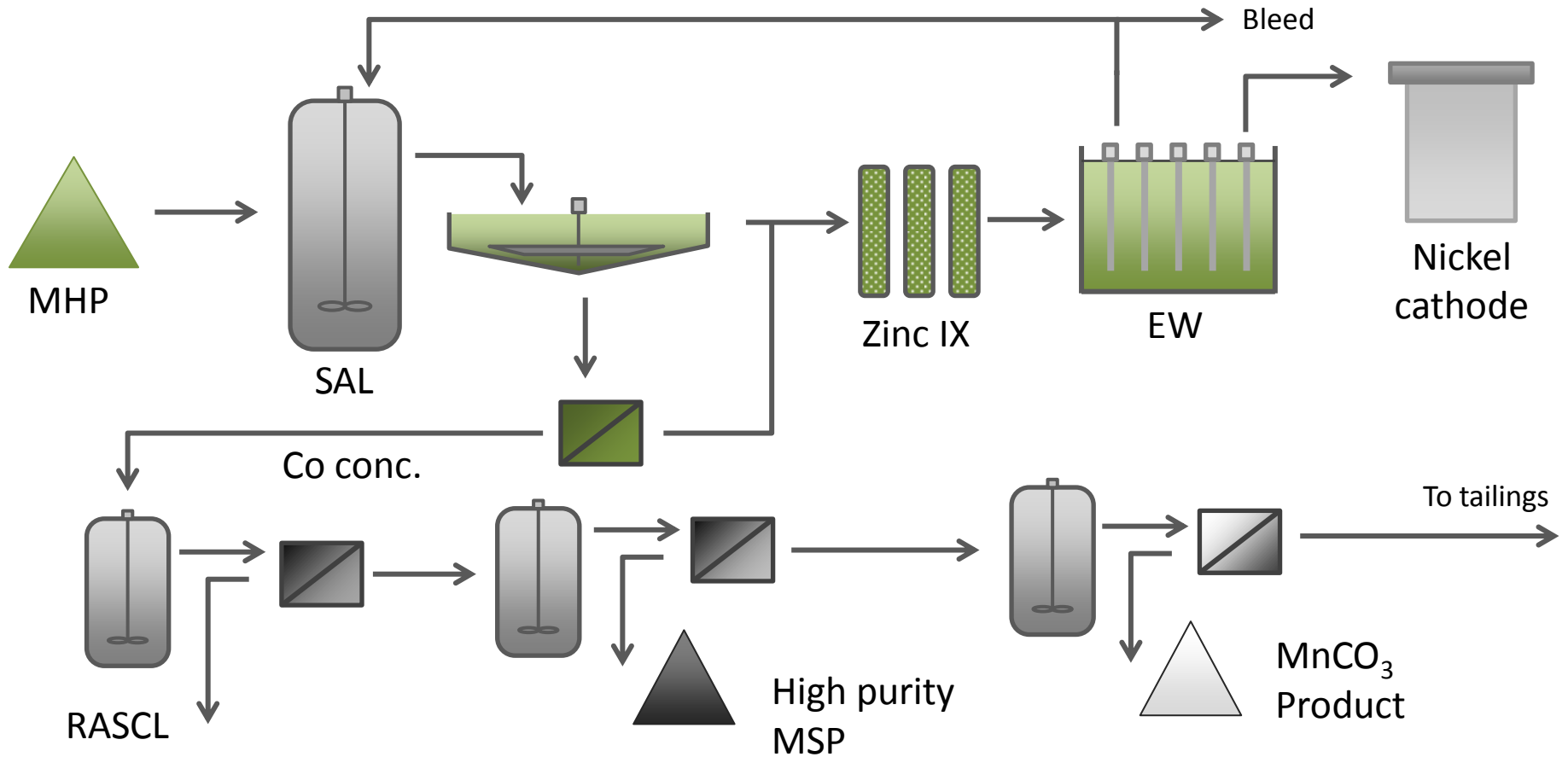
Single step leach and cobalt-nickel separation



Demonstrated and moving toward integrated pilot

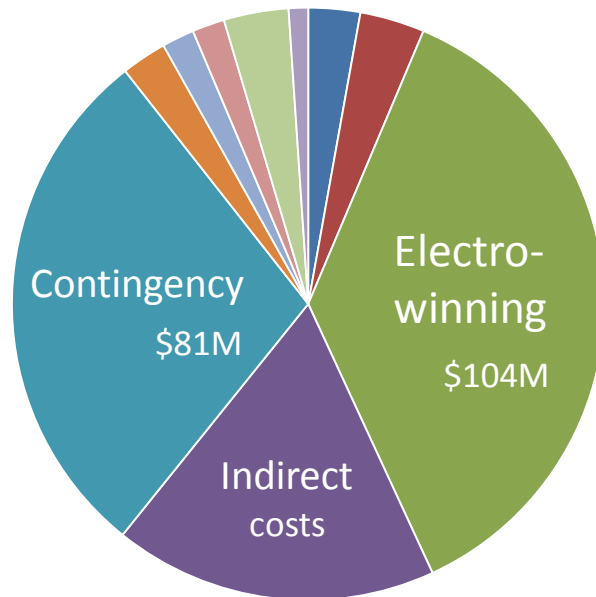


Completed flow sheet development

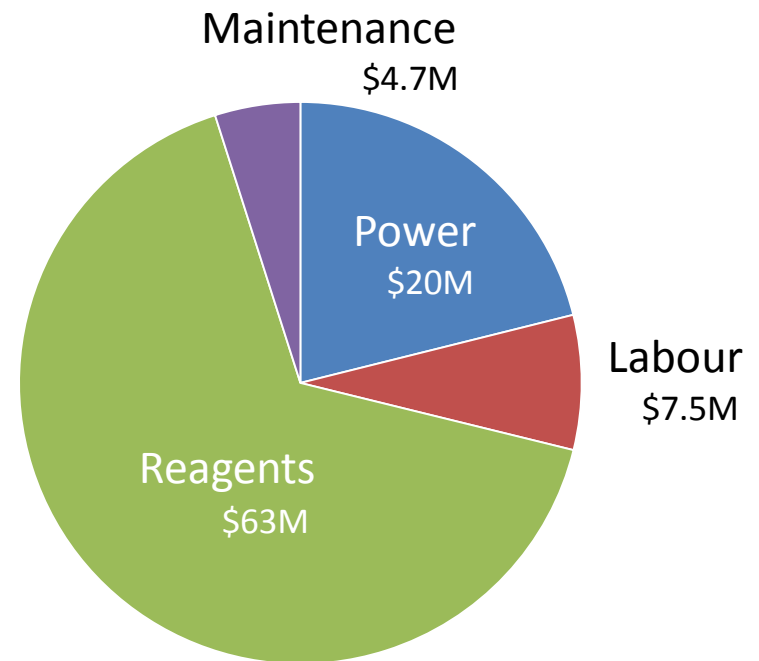


Concept-level study by Ausenco

- 50,000 tpa of nickel as MHP in feed
- $\pm 50\%$ confidence for cost estimates (AUD)
- Plant built on existing Australian site



\$284M Capital Cost



\$96M Annual Operating Cost

Significantly improved profitability over MHP

	MHP	MHP + SAL	SX
Mine life	20 years		
Annual capacity	50,000 tpa Ni and 2,500 tpa Co		
Annual revenue ¹	\$745M	\$1,060M	\$1,075M
Cash costs	\$6 / lb of Ni ²	\$6.87 / lb of Ni ³	\$5 / lb of Ni ⁴
Annual operating costs	\$660M	\$757M	\$550M
Annual gross profit	\$82M	\$301M	\$523M
Capital cost	\$1,500M	\$1,784M	\$3,500M
Payback period	19 years	6 years	7 years
NPV (8%)	-\$690M	\$1,107M	\$1,528M
IRR	1%	16%	14%
Ni price for +’ve NPV(8%)⁵	\$9.85	\$8.00	\$7.70

1. Long term metal prices of \$9/lb for nickel, and \$15/lb for cobalt
2. C1 cash costs for MHP production approximated from First Quantum’s Ravensthorpe operation
3. C1 cash costs for MHP+SAL approximated from Ravensthorpe plus SAL cost estimate by Ausenco
4. C1 cash costs for SX approximated from multiple sources (Brooks Hunt, Minecost)
5. Assuming no change in cash costs

Opportunity to stage capital expenses

- Direct shipping of high grade ore
- Producing MHP as next product has advantages
 - Lower initial capital expense
 - Faster ramp-up
 - Lower technical risk by decoupling front and back-end
 - Potentially finance back-end from MHP revenue
- Build backend to upgrade MHP to high value products
 - A number of options
 - New low capital cost, low complexity selective acid leach option

Opportunity for a MHP toll refinery

	MHP Refinery
Plant life	20 years
Annual capacity	50,000 tpa of Ni 2,500 tpa of Co
Annual revenue ¹	\$1,060M
Cash costs	\$0.87 / lb of Ni ²
Annual MHP purchase costs ³	\$744M
Total annual costs	\$840M
Annual gross profit	\$218M
Capital cost	\$284M
Payback period	2 years
NPV (8%)	\$1,812M
IRR	77%

1. Long term metal prices of \$9/lb for nickel, and \$15/lb for cobalt
2. Cash costs do not include cobalt credits
3. Assuming MHP purchase at 75% contained nickel value

- Even lower capital cost options utilising existing nickel refinery:
 - Existing nickel matte refinery (e.g. Sherritt-Gordon process)
 - Existing ferronickel smelter (creating an NHP feed)

Summary

- Nickel laterite resource owners have multiple options
 - Direct shipping ore, low capital cost but low product value, especially for low grade ore
 - MHP production, reasonable capital cost but low product value (70-75%)
 - Metal production by SX or FeNi, very high capital cost but high product value (100%)
- MHP is a popular option, but low payable makes it uneconomic at low nickel prices
- High value products are key to being economic at low nickel prices
- But, need a low capital and operating cost to get shareholder return

Thank you

- Metallica Minerals Ltd, BHP Billiton Nickel West, First Quantum Minerals Ltd, Highlands Pacific Ltd, Teck Resources Ltd, Queensland Nickel
- Boyd Willis, David White, Chad Czerny